x: 4

2/2/26

VERIFICATION OF THEVENIN'S THEOREM

AIM:

To verify the Thevenin's theorem practically and theorectically for the given DC circuit.

APPARATUS REQUIRED:

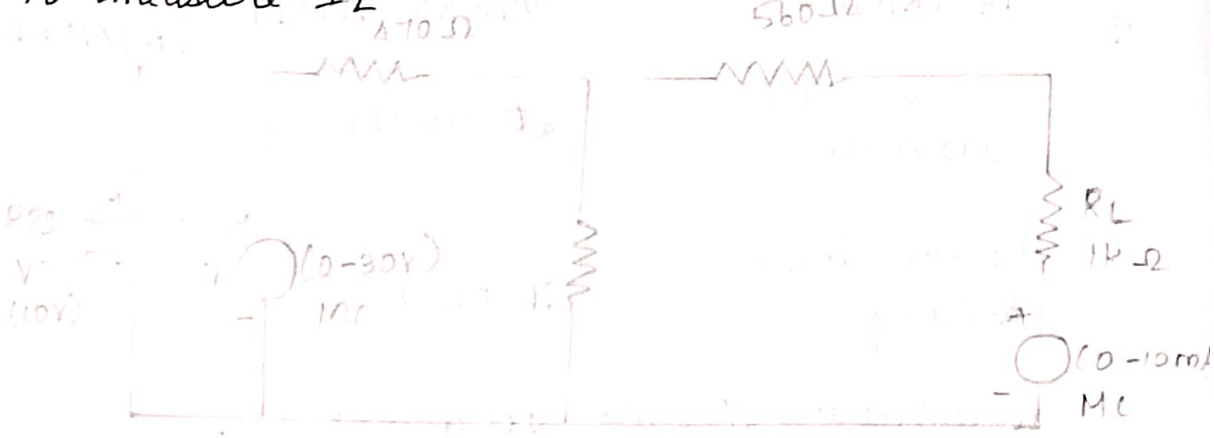
S.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power Supply	(0-30 V)	1
2.	Voltmeter	(0-30 V) MC	1
3.	Ammeter	(0-10 mA) MC	1
4.	Resistors	470 Ω, 560 Ω, 1 kΩ	2, 1, 1
5.	Bread board	–	1
6.	Multimeter	–	1

PROCEDURE:

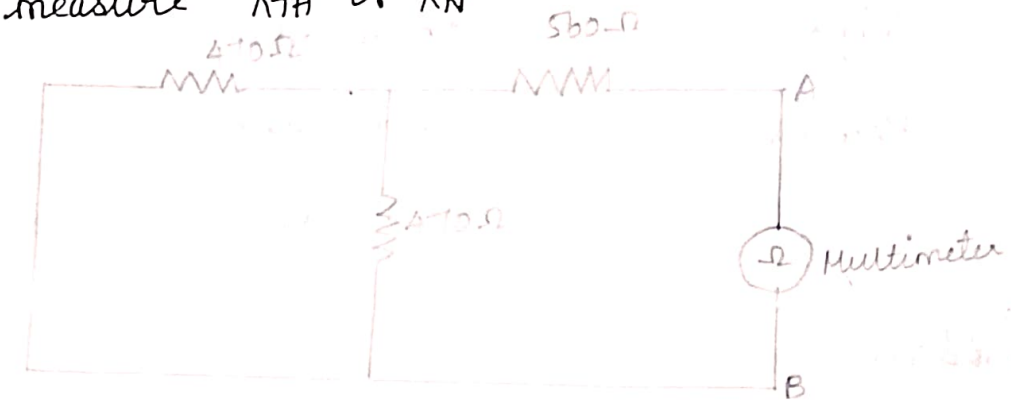
* Make the connections as per the Circuit Diagram 1.
* Vary the RPS and set an input voltage of 10V.

CIRCUIT DIAGRAM : VERIFICATION OF THEVENIN'S THEOREM

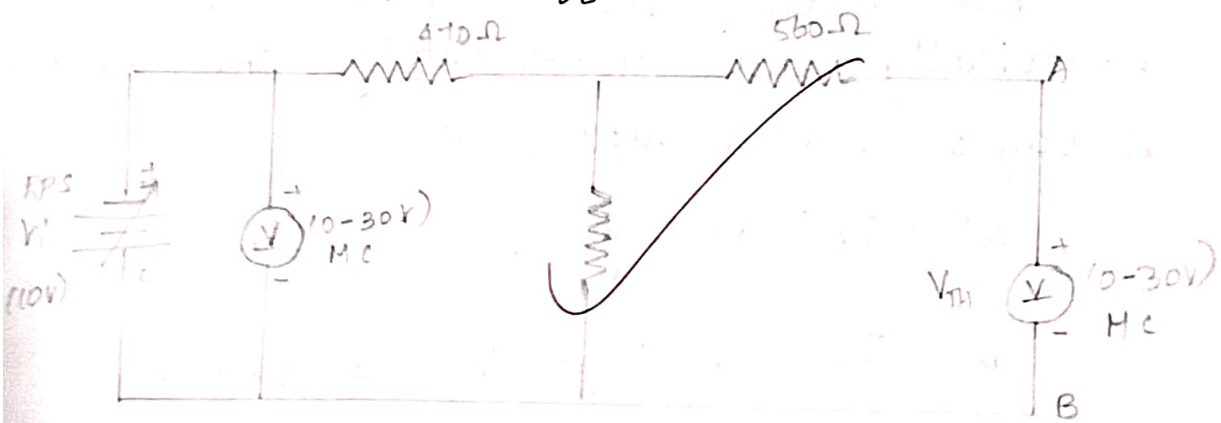
To measure I_L



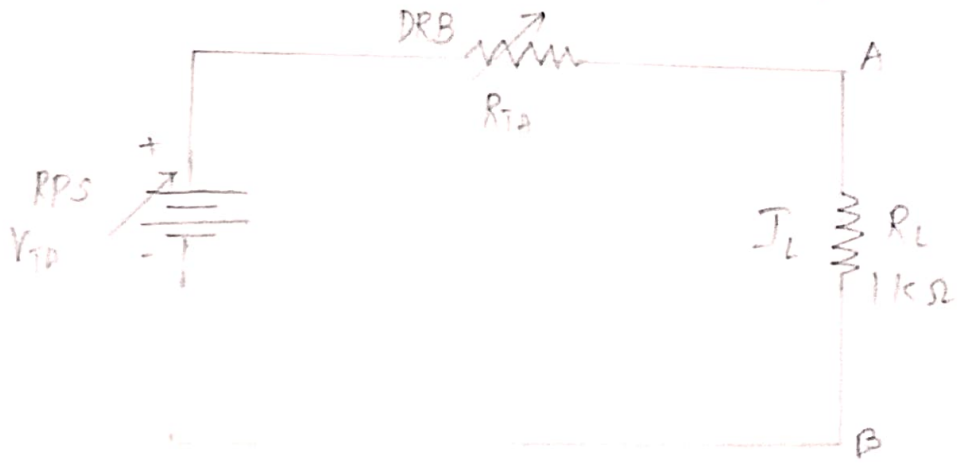
To measure R_{TH} or R_N



To measure V_{TH} or V_{OC}



Thevenin's equivalent circuit



TABULAR COLUMN : 1

To measure I_L

V_i (volts)	I_L (amps)
10V	25mA

TABULAR COLUMN : 2

To measure R_{TH} or R_N

From circuit diagram 2, $R_{Th} = R_N = 7898\,\Omega$

R_N (Ω)	R_{TH} (Ω)

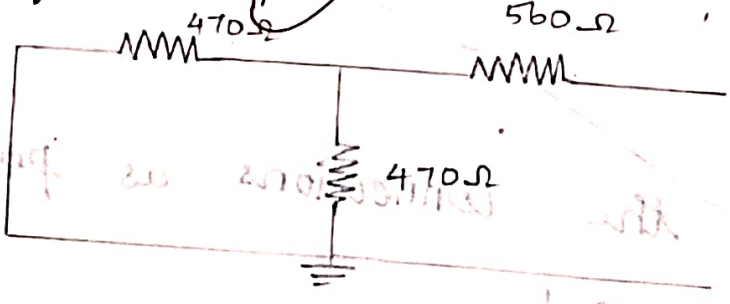
TABULAR COLUMN : 3

To measure V_{Th} or V_{oc}

V_i (volts)	V_{Th} (volts)
10V	4.5V

CALCULATION :

To find R_{TH},



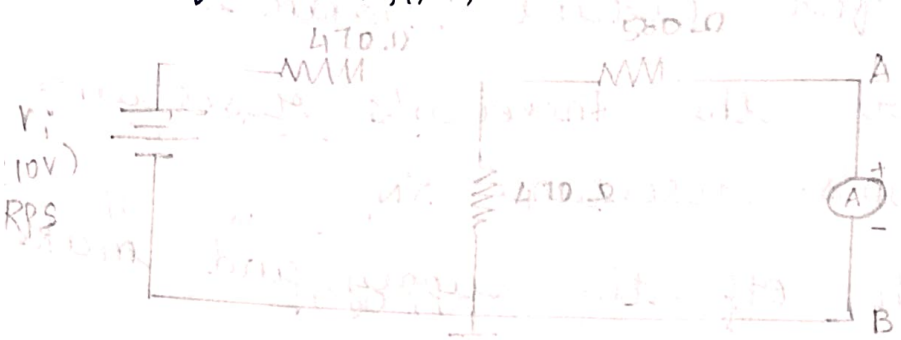
$$R_P = \frac{470 \times 470}{470 + 470}$$ [470 Ω and 470 Ω are in parallel]

$= 235\,\Omega$

$R_{TH} = 235 + 560$

$$\boxed{R_{TH} = 795\Omega}$$

To find V_{TH},



$$V_{TH} = \frac{V_s \times R_2}{R_1 + R_2}$$

$$= \frac{10 \times 470}{470 + 470}$$

$$= \frac{10}{2}$$

$$\boxed{V_{TH} = 5V}$$

To find I_L,

$$I_L = \frac{V_{TH}}{R_L + R_{TH}}$$

$$= \frac{5}{1000 + 795}$$

$$= \frac{5}{1795}$$

$$\boxed{I_L = 2.7\,mA}$$

* Note down the voltmeter reading (V_i) and ammeter reading (I_L) in Tabular Column 1.

* Switch off the supply and make connections for Circuit Diagram 2.

* Measure the Thevenin's resistance R_{TH} = Norton's resistance R_N.

* Switch off the supply and make connections for Circuit diagram: 3.

* Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and V_{TH} ($= V_{OC}$) in Tabular Column:

* Switch off the supply and make connections for Circuit Diagram 4.

* Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading I_N ($= I_{SC}$) in Tabular Column 4.

* Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagram 5 & 6 respectively

* Calculate the I_L value using the formula,

Thevenin's Theorem

$$I_L = V_{TH} / (R_{TH} + R_L)$$

Norton's theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

* Theorectically verify the Norton's theorem.

MARKS SPLIT UP :

S. NO	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental / setup / connections	20 marks	20
2.	Observation / Readings	20 marks	20
3.	Calculations / Results	20 marks	20
4.	Viva	20 marks	20
5.	Record	20 marks	20
		Total	100

RESULT :

13/12/2026 Thus the Thevenin's theorem is verified both practically and theorectically.